

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Logarithmic Trigonometric Substitution

Evaluate the definite integral:

$$\int_{\pi/4}^{\pi/3} \frac{\cot(x) + \tan(x)}{\ln^2(\tan(x)) + 1 + \ln(\sec(x) - 1) + \ln(\sec(x) + 1)} dx$$

Solution: First, simplify the numerator and the denominator of the integrand. For the numerator, express the trigonometric functions in terms of sine and cosine:

$$\cot(x) + \tan(x) = \frac{\cos(x)}{\sin(x)} + \frac{\sin(x)}{\cos(x)} = \frac{\cos^2(x) + \sin^2(x)}{\sin(x) \cos(x)} = \frac{1}{\sin(x) \cos(x)}$$

For the denominator, combine the logarithmic terms involving secant using the property $\ln(a) + \ln(b) = \ln(ab)$:

$$\ln(\sec(x) - 1) + \ln(\sec(x) + 1) = \ln((\sec(x) - 1)(\sec(x) + 1)) = \ln(\sec^2(x) - 1)$$

Using the Pythagorean identity $\sec^2(x) - 1 = \tan^2(x)$, this becomes $\ln(\tan^2(x))$, which can be written as $2 \ln(\tan(x))$. Thus, the entire denominator simplifies to:

$$\ln^2(\tan(x)) + 2 \ln(\tan(x)) + 1 = (\ln(\tan(x)) + 1)^2$$

The integral I now takes the much simpler form:

$$I = \int_{\pi/4}^{\pi/3} \frac{1}{\sin(x) \cos(x) (\ln(\tan(x)) + 1)^2} dx$$

Apply the substitution $u = \ln(\tan(x))$. Differentiating with respect to x yields:

$$du = \frac{1}{\tan(x)} \sec^2(x) dx = \frac{\cos(x)}{\sin(x)} \frac{1}{\cos^2(x)} dx = \frac{1}{\sin(x) \cos(x)} dx$$

This perfectly matches the remaining term in our numerator. Next, update the limits of integration: When $x = \pi/4$, $u = \ln(\tan(\pi/4)) = \ln(1) = 0$. When $x = \pi/3$, $u = \ln(\tan(\pi/3)) = \ln(\sqrt{3}) = \frac{1}{2} \ln(3)$. Substituting these into the integral gives:

$$I = \int_0^{\frac{1}{2} \ln(3)} \frac{du}{(u + 1)^2}$$

Evaluate this standard rational integral:

$$I = \left[-\frac{1}{u + 1} \right]_0^{\frac{1}{2} \ln(3)} = -\frac{1}{\frac{1}{2} \ln(3) + 1} - \left(-\frac{1}{0 + 1} \right) = 1 - \frac{2}{\ln(3) + 2}$$

Combining into a single fraction:

$$I = \frac{\ln(3) + 2 - 2}{\ln(3) + 2} = \frac{\ln(3)}{\ln(3) + 2}$$

1.2 Hard Integral

Infinite Logarithmic Sine Integral

Evaluate the integral:

$$\int_0^{\infty} \frac{\sin(\ln(x))}{1-x^2} dx$$

Solution: Let the given integral be I . We first split the domain of integration into two parts: $[0, 1]$ and $[1, \infty)$:

$$I = \int_0^1 \frac{\sin(\ln(x))}{1-x^2} dx + \int_1^{\infty} \frac{\sin(\ln(x))}{1-x^2} dx$$

Consider the second integral. Apply the substitution $x = \frac{1}{u}$, which gives $dx = -\frac{1}{u^2} du$. The limits change from $[1, \infty)$ to $[1, 0]$:

$$\int_1^{\infty} \frac{\sin(\ln(x))}{1-x^2} dx = \int_1^0 \frac{\sin(\ln(1/u))}{1-(1/u)^2} \left(-\frac{1}{u^2}\right) du$$

Using the property $\ln(1/u) = -\ln(u)$ and the odd nature of the sine function $\sin(-\ln(u)) = -\sin(\ln(u))$, the integral becomes:

$$\int_0^1 \frac{-\sin(\ln(u))}{\frac{u^2-1}{u^2}} \frac{1}{u^2} du = \int_0^1 \frac{-\sin(\ln(u))}{u^2-1} du = \int_0^1 \frac{\sin(\ln(u))}{1-u^2} du$$

Replacing the dummy variable u with x , we see this is identical to the first integral. Thus:

$$I = 2 \int_0^1 \frac{\sin(\ln(x))}{1-x^2} dx$$

Next, substitute $t = -\ln(x)$, which means $x = e^{-t}$ and $dx = -e^{-t} dt$. As $x \rightarrow 0$, $t \rightarrow \infty$; as $x \rightarrow 1$, $t \rightarrow 0$.

$$I = 2 \int_{\infty}^0 \frac{\sin(-t)}{1-e^{-2t}} (-e^{-t}) dt = -2 \int_0^{\infty} \frac{e^{-t} \sin(t)}{1-e^{-2t}} dt$$

We can expand $\frac{1}{1-e^{-2t}}$ into a geometric series since $e^{-2t} < 1$ for $t > 0$:

$$\frac{1}{1-e^{-2t}} = \sum_{n=0}^{\infty} e^{-2nt}$$

Substitute this series back into the integral and swap the integral and sum:

$$I = -2 \sum_{n=0}^{\infty} \int_0^{\infty} e^{-t} e^{-2nt} \sin(t) dt = -2 \sum_{n=0}^{\infty} \int_0^{\infty} e^{-(2n+1)t} \sin(t) dt$$

The integral $\int_0^{\infty} e^{-at} \sin(t) dt$ is a known standard integral that evaluates to $\frac{1}{a^2+1}$. Here, $a = 2n+1$:

$$I = -2 \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2+1}$$

This infinite series is the sum over all positive odd integers of $\frac{1}{k^2+1}$. It is known from the Mittag-Leffler expansion of the hyperbolic cotangent that:

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2+1} = \frac{\pi}{4} \tanh\left(\frac{\pi}{2}\right)$$

Therefore, substituting this back gives:

$$I = -2 \left(\frac{\pi}{4} \tanh\left(\frac{\pi}{2}\right) \right) = -\frac{\pi}{2} \tanh\left(\frac{\pi}{2}\right)$$

Differentials

2.1 Medium Differential

Coupled Trigonometric ODE

A curve $y(x)$ passing through the origin satisfies the differential equation:

$$\frac{dy}{dx} = \cos(x)(\cos(y) - \sin(y)) - \sin(x)(\cos(y) + \sin(y))$$

Evaluate: $\cot\left(\frac{y(\ln(2)) + \ln(2)}{2}\right)$

Solution: First, expand and collect terms on the right-hand side of the differential equation:

$$\frac{dy}{dx} = \cos(x)\cos(y) - \cos(x)\sin(y) - \sin(x)\cos(y) - \sin(x)\sin(y)$$

Group the terms using the trigonometric addition formulas:

$$\frac{dy}{dx} = (\cos(x)\cos(y) - \sin(x)\sin(y)) - (\sin(x)\cos(y) + \cos(x)\sin(y))$$

$$\frac{dy}{dx} = \cos(x+y) - \sin(x+y)$$

Use the substitution $u = x + y$, so $\frac{du}{dx} = 1 + \frac{dy}{dx}$. Substituting this in:

$$\frac{du}{dx} - 1 = \cos(u) - \sin(u) \implies \frac{du}{dx} = 1 + \cos(u) - \sin(u)$$

This is a separable differential equation:

$$\int \frac{du}{1 + \cos(u) - \sin(u)} = \int dx$$

To solve the integral on the left, use the Weierstrass substitution (tangent half-angle), $t = \tan(u/2)$. Then $du = \frac{2}{1+t^2}dt$, $\cos(u) = \frac{1-t^2}{1+t^2}$, and $\sin(u) = \frac{2t}{1+t^2}$:

$$1 + \cos(u) - \sin(u) = 1 + \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} = \frac{2-2t}{1+t^2}$$

The integral becomes:

$$\int \frac{1}{\frac{2-2t}{1+t^2}} \frac{2}{1+t^2} dt = \int \frac{1}{1-t} dt = -\ln|1-t|$$

Equating this to the right side gives:

$$-\ln\left|1 - \tan\left(\frac{u}{2}\right)\right| = x + C$$

Since the curve passes through the origin, $y(0) = 0$, so $u(0) = 0$. Using this initial condition to find C :

$$-\ln|1 - \tan(0)| = 0 + C \implies C = 0$$

The equation describing the curve is therefore:

$$-\ln\left|1 - \tan\left(\frac{x+y}{2}\right)\right| = x \implies 1 - \tan\left(\frac{x+y}{2}\right) = e^{-x}$$

Solving for the tangent term:

$$\tan\left(\frac{x+y}{2}\right) = 1 - e^{-x}$$

We are asked to evaluate $\cot\left(\frac{y(\ln(2)) + \ln(2)}{2}\right)$. This corresponds exactly to evaluating the reciprocal of the tangent expression when $x = \ln(2)$:

$$\tan\left(\frac{\ln(2) + y(\ln(2))}{2}\right) = 1 - e^{-\ln(2)} = 1 - \frac{1}{2} = \frac{1}{2}$$

Since cotangent is the reciprocal of tangent:

$$\cot\left(\frac{y(\ln(2)) + \ln(2)}{2}\right) = \frac{1}{1/2} = 2$$

2.2 Hard Differential

Chebyshev Equation Variant

Solve the second-order differential equation:

$$(1 - x^2)y'' - xy' + 4y = 0$$

Given the initial conditions $y(0) = 1.5$ and $y'(0) = 3^{1.5}$, find the exact value of $y(0.5)$.

Solution: This is a standard Chebyshev differential equation of the form $(1 - x^2)y'' - xy' + n^2y = 0$, where $n = 2$. We solve this by applying the trigonometric substitution $x = \sin(\theta)$, or $\theta = \arcsin(x)$. Using the chain rule to transform the derivatives with respect to x into derivatives with respect to θ :

$$\frac{dy}{dx} = \frac{dy}{d\theta} \frac{d\theta}{dx} = \frac{dy}{d\theta} \frac{1}{\cos(\theta)}$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta} \left(\frac{dy}{d\theta} \frac{1}{\cos(\theta)} \right) \frac{1}{\cos(\theta)} = \left(\frac{d^2y}{d\theta^2} \frac{1}{\cos(\theta)} + \frac{dy}{d\theta} \frac{\sin(\theta)}{\cos^2(\theta)} \right) \frac{1}{\cos(\theta)}$$

$$\frac{d^2y}{dx^2} = \frac{1}{\cos^2(\theta)} \frac{d^2y}{d\theta^2} + \frac{\sin(\theta)}{\cos^3(\theta)} \frac{dy}{d\theta}$$

Substitute these into the original differential equation, noting that $1 - x^2 = \cos^2(\theta)$ and $x = \sin(\theta)$:

$$\cos^2(\theta) \left(\frac{1}{\cos^2(\theta)} y''_{\theta} + \frac{\sin(\theta)}{\cos^3(\theta)} y'_{\theta} \right) - \sin(\theta) \left(\frac{1}{\cos(\theta)} y'_{\theta} \right) + 4y = 0$$

$$y''_{\theta} + \tan(\theta) y'_{\theta} - \tan(\theta) y'_{\theta} + 4y = 0 \implies y''_{\theta} + 4y = 0$$

This is a simple harmonic oscillator equation with the general solution:

$$y(\theta) = C_1 \cos(2\theta) + C_2 \sin(2\theta)$$

Re-substituting $x = \sin(\theta)$, we can express $\cos(2\theta)$ and $\sin(2\theta)$ in terms of x :

$$\cos(2\theta) = 1 - 2\sin^2(\theta) = 1 - 2x^2$$

$$\sin(2\theta) = 2\sin(\theta)\cos(\theta) = 2x\sqrt{1-x^2}$$

The general solution in terms of x is:

$$y(x) = C_1(1 - 2x^2) + C_2(2x\sqrt{1-x^2})$$

Apply the initial conditions to find C_1 and C_2 . First, for $y(0) = 1.5$:

$$y(0) = C_1(1 - 0) + C_2(0) = C_1 = 1.5$$

Next, find $y'(x)$:

$$y'(x) = -4C_1x + 2C_2\sqrt{1-x^2} - \frac{2C_2x^2}{\sqrt{1-x^2}}$$

Evaluate at $x = 0$ using $y'(0) = 3^{1.5} = 3\sqrt{3}$:

$$y'(0) = 0 + 2C_2(1) - 0 = 2C_2 = 3\sqrt{3} \implies C_2 = \frac{3\sqrt{3}}{2}$$

The specific solution is:

$$y(x) = 1.5(1 - 2x^2) + \frac{3\sqrt{3}}{2}(2x\sqrt{1-x^2}) = 1.5(1 - 2x^2) + 3\sqrt{3}x\sqrt{1-x^2}$$

Finally, evaluate this function at $x = 0.5$:

$$y(0.5) = 1.5(1 - 2(0.5)^2) + 3\sqrt{3}(0.5)\sqrt{1 - (0.5)^2} = 1.5(1 - 0.5) + \frac{3\sqrt{3}}{2}\sqrt{0.75}$$

$$y(0.5) = 1.5(0.5) + \frac{3\sqrt{3}}{2} \frac{\sqrt{3}}{2} = 0.75 + \frac{9}{4} = 0.75 + 2.25 = 3$$

Derivatives

3.1 Medium Derivative

Leibniz Integral Rule

Let $f(x) = x \int_{\sqrt[3]{x}}^x \frac{dt}{1+t^3}$. Find $f'(1)$.

Solution: To find the derivative of $f(x)$, we must apply the product rule and the Leibniz integral rule. Let $f(x) = g(x) \cdot h(x)$, where $g(x) = x$ and $h(x) = \int_{\sqrt[3]{x}}^x \frac{dt}{1+t^3}$. The product rule gives:

$$f'(x) = g'(x)h(x) + g(x)h'(x) = \left(1 \cdot \int_{\sqrt[3]{x}}^x \frac{dt}{1+t^3}\right) + x \frac{d}{dx} \left(\int_{\sqrt[3]{x}}^x \frac{dt}{1+t^3}\right)$$

Applying the Leibniz integral rule to differentiate the integral $h(x)$ with variable bounds:

$$h'(x) = \frac{1}{1+x^3} \frac{d}{dx}(x) - \frac{1}{1+(\sqrt[3]{x})^3} \frac{d}{dx}(\sqrt[3]{x})$$

$$h'(x) = \frac{1}{1+x^3}(1) - \frac{1}{1+x} \left(\frac{1}{3}x^{-2/3}\right) = \frac{1}{1+x^3} - \frac{1}{3x^{2/3}(1+x)}$$

We now assemble $f'(x)$:

$$f'(x) = \left(\int_{\sqrt[3]{x}}^x \frac{dt}{1+t^3}\right) + x \left(\frac{1}{1+x^3} - \frac{1}{3x^{2/3}(1+x)}\right)$$

To find $f'(1)$, we evaluate this expression at $x = 1$. First, look at the integral component:

$$\int_{\sqrt[3]{1}}^1 \frac{dt}{1+t^3} = \int_1^1 \frac{dt}{1+t^3} = 0$$

Next, evaluate the second part:

$$f'(1) = 0 + 1 \cdot \left(\frac{1}{1+1^3} - \frac{1}{3(1)^{2/3}(1+1)}\right) = \frac{1}{2} - \frac{1}{3(2)} = \frac{1}{2} - \frac{1}{6}$$

$$f'(1) = \frac{3}{6} - \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

3.2 Hard Derivative

Limit and Inverse Function Derivative

Compute $(f^{-1}(16))'$ if

$$f(x) = \lim_{n \rightarrow \infty} \left(\sqrt[n]{x^n + \ln^n(x)} \right)^{\sqrt[n]{2^{xn} + (\tan^{-1}(x))^n}}$$

Solution: First, evaluate the limit to define $f(x)$ for values of x where $f(x)$ is around 16. We analyze the two limits separately: the base and the exponent. For the base, since $x > \ln(x)$ for $x > 0$:

$$\lim_{n \rightarrow \infty} \sqrt[n]{x^n + \ln^n(x)} = \lim_{n \rightarrow \infty} x \sqrt[n]{1 + \left(\frac{\ln(x)}{x}\right)^n} = x(1) = x$$

For the exponent, note that $2^x > \tan^{-1}(x)$ for all $x > 0$. The dominant term is $2^{xn} = (2^x)^n$:

$$\lim_{n \rightarrow \infty} \sqrt[n]{2^{xn} + (\tan^{-1}(x))^n} = \lim_{n \rightarrow \infty} 2^x \sqrt[n]{1 + \left(\frac{\tan^{-1}(x)}{2^x}\right)^n} = 2^x(1) = 2^x$$

Therefore, the function drastically simplifies to:

$$f(x) = x^{2^x}$$

We need to find $(f^{-1}(16))'$. By the inverse function theorem, $(f^{-1}(y))' = \frac{1}{f'(x)}$ where $y = f(x)$. We must find the x such that $f(x) = 16$:

$$x^{2^x} = 16$$

By inspection, testing small integers, we see that if $x = 2$, then $2^{2^2} = 2^4 = 16$. So, $f^{-1}(16) = 2$. Next, find the derivative $f'(x)$ using logarithmic differentiation:

$$\ln(f(x)) = 2^x \ln(x)$$

Differentiate both sides with respect to x :

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} (2^x \ln(x)) = 2^x \ln(2) \ln(x) + \frac{2^x}{x}$$

Solve for $f'(x)$:

$$f'(x) = x^{2^x} \left(2^x \ln(2) \ln(x) + \frac{2^x}{x} \right)$$

Evaluate $f'(2)$:

$$f'(2) = 2^{2^2} \left(2^2 \ln(2) \ln(2) + \frac{2^2}{2} \right) = 16 \left(4(\ln(2))^2 + 2 \right) = 64(\ln(2))^2 + 32$$

Finally, compute the derivative of the inverse function at 16:

$$(f^{-1}(16))' = \frac{1}{f'(2)} = \frac{1}{32 + 64(\ln(2))^2}$$

Physics & Engineering

4.1 Mechanics

Constant Acceleration Distance

A car accelerates from rest to a speed of $v = 67 \text{ m s}^{-1}$ in a time interval of $t = 4.1 \text{ s}$. Assuming the acceleration is constant, calculate the net distance traveled in meters. Round your final answer to two decimal places. Use two significant figures for intermediate calculations.

Solution: When a vehicle travels under constant acceleration, its average velocity over the time interval is the arithmetic mean of its initial and final velocities. The total distance traveled d is the product of the average velocity and the total time:

$$d = \left(\frac{v_0 + v}{2} \right) t$$

Given the problem conditions, the car starts from rest, so its initial velocity $v_0 = 0$. Its final velocity is $v = 67 \text{ m s}^{-1}$, and the time taken is $t = 4.1 \text{ s}$. Substitute these values into the kinematic equation:

$$d = \left(\frac{0 + 67}{2} \right) \times 4.1$$

$$d = 33.5 \times 4.1$$

Carrying out the multiplication:

$$d = 137.35 \text{ m}$$

The value is exact to two decimal places without any truncation or rounding needed. Thus, the net distance traveled is 137.35 m.

4.2 Quantum Mechanics

Particle Confinement Kinetic Energy

An electron is confined within a region of width $\Delta x = 2.50 \times 10^{-14} \text{ m}$. Estimate the minimum kinetic energy of the electron in MeV. Use the following constants: $\hbar = 1.055 \times 10^{-34} \text{ J s}$, $m_e = 9.109 \times 10^{-31} \text{ kg}$ and $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$.

Solution: According to the Heisenberg Uncertainty Principle, the minimum uncertainty in momentum Δp for a given positional confinement Δx is:

$$\Delta x \Delta p \geq \frac{\hbar}{2} \implies \Delta p_{\min} = \frac{\hbar}{2\Delta x}$$

Substitute the given values into the formula to find the minimum momentum:

$$\Delta p_{\min} = \frac{1.055 \times 10^{-34}}{2 \times (2.50 \times 10^{-14})} = \frac{1.055 \times 10^{-34}}{5.00 \times 10^{-14}} = 2.11 \times 10^{-21} \text{ kg m s}^{-1}$$

The kinetic energy K of a non-relativistic particle can be expressed in terms of its momentum as $K = \frac{p^2}{2m}$. Setting $p = \Delta p_{min}$ gives the minimum estimated kinetic energy:

$$K_{min} = \frac{(\Delta p_{min})^2}{2m_e} = \frac{(2.11 \times 10^{-21})^2}{2 \times (9.109 \times 10^{-31})}$$

$$K_{min} = \frac{4.4521 \times 10^{-42}}{1.8218 \times 10^{-30}} \approx 2.4438 \times 10^{-12} \text{ J}$$

Convert the kinetic energy from Joules to electron-volts (eV) using the provided conversion factor:

$$K_{min}(\text{in eV}) = \frac{2.4438 \times 10^{-12} \text{ J}}{1.602 \times 10^{-19} \text{ J/eV}} \approx 1.525 \times 10^7 \text{ eV}$$

Finally, convert from eV to mega-electron-volts (MeV) by dividing by 10^6 :

$$K_{min} = 15.25 \text{ MeV}$$

4.3 Electromagnetism

Induction in a Moving Loop

A rectangular loop of sides 12 cm and 5 cm, with its sides parallel to the x -axis and y -axis respectively, moves with a velocity of 5 cm s^{-1} in the positive x direction, in a space containing a variable magnetic field in the positive z direction. The field has a gradient of $10^{-3} \text{ T cm}^{-1}$ along the negative x direction and it is decreasing with time at the rate of 10^{-3} T s^{-1} . If the resistance of the loop is $6 \text{ m}\Omega$, find the power dissipated by the loop as heat in units of 10^{-9} W .

Solution: First, convert all units to standard SI units: Area of the loop: $A = (0.12 \text{ m}) \times (0.05 \text{ m}) = 0.006 \text{ m}^2$ Velocity of the loop: $v = 5 \text{ cm s}^{-1} = 0.05 \text{ m s}^{-1}$ Magnetic field gradient: $\frac{\partial B}{\partial x} = -10^{-3} \text{ T cm}^{-1} = -0.1 \text{ T m}^{-1}$ (Note: the gradient is positive in the negative x direction, so it decreases as x increases) Time rate of change of B : $\frac{\partial B}{\partial t} = -10^{-3} \text{ T s}^{-1}$ Resistance: $R = 6 \text{ m}\Omega = 6 \times 10^{-3} \Omega$

The total rate of change of the magnetic field experienced by the moving loop is described by the convective derivative:

$$\frac{dB}{dt}_{total} = \frac{\partial B}{\partial t} + v \frac{\partial B}{\partial x}$$

Substitute the known values to find the total rate of change:

$$\frac{dB}{dt}_{total} = -10^{-3} + (0.05)(-0.1) = -0.001 - 0.005 = -0.006 \text{ T s}^{-1}$$

According to Faraday's Law, the induced electromotive force (EMF), ϵ , is given by the negative rate of change of magnetic flux:

$$\epsilon = -\frac{d\Phi}{dt} = -A \frac{dB}{dt}_{total}$$

$$\epsilon = -(0.006 \text{ m}^2)(-0.006 \text{ T s}^{-1}) = 3.6 \times 10^{-5} \text{ V}$$

The power P dissipated as heat due to the induced current can be found using the formula $P = \epsilon^2/R$:

$$P = \frac{(3.6 \times 10^{-5})^2}{6 \times 10^{-3}} = \frac{1.296 \times 10^{-9}}{6 \times 10^{-3}} = 2.16 \times 10^{-7} \text{ W}$$

To express this in units of 10^{-9} W , adjust the scientific notation:

$$P = 216 \times 10^{-9} \text{ W}$$

The power dissipated by the loop is 216 units of 10^{-9} W .